

GENERAL THETA FOUNDATIONS I: SHARP WIDTH OF HIDDEN ROTATION EXPERIMENTS

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ABSTRACT. We determine the order of the least hidden-state width of a uniformly positive binary-command rotation experiment. For every badly approximable angle and every fixed sufficiently small uniform output error, the minimum width is $\Theta(N^{1/3})$. The lower bound permits arbitrary stochastic transitions, cut-dependent state spaces, unobservable hidden directions, and a different machine for every horizon. It therefore matches the previous resonant-polygon construction without a vector-state restriction. The proof groups commands into independent packets whose rotation counts are separated. Compressing one packet into k labels loses a fixed multiple of k^{-2} of the conditional phase amplitude, while the packet uses $2k - 1$ commands. Multiplying these losses yields both the sharp order and a nonuniform occupation bound. A representation-valued version gives an orbit-quantization budget for general compact groups; for the separate model with all $\mathrm{SO}(d)$ rotations available as commands it yields the matched order $N^{(d-1)/2}$. We also bound the actual optimal terminal simulation error, independently of cancellation in local deficiency certificates. Throughout, width counts available persistent labels in a clocked atomic stochastic-row model; descriptions, arithmetic and exact real sampling are not charged. The fixed-alphabet rotation theorem and the compact-command extension have distinct input conventions.

1. INTRODUCTION

A small sufficient state at each separate experimental cut need not be compatible with small states at all cuts of one stochastic machine. The rotation experiment below makes this discrepancy quantitative while keeping the seed, command, query and answer alphabets fixed. Its separate positive minimum is three at every cut. Its simultaneous minimum grows with the number of commands, even when a fixed uniform approximation error is allowed.

Write R_α for planar rotation through $2\pi\alpha$. A seed $x \in \{-1, 1\}^2$ is followed by N binary commands and one coordinate query $j \in \{1, 2\}$. The answer $B \in \{-1, 1\}$ must have mean

$$(1) \quad \mathbb{E}[B \mid x, c_1, \dots, c_N, j] = \frac{1}{10} e_j^T R_\alpha^{c_1 + \dots + c_N} x.$$

All answer probabilities belong to $[(1 - \sqrt{2}/10)/2, (1 + \sqrt{2}/10)/2]$, independently of N . Let $W_{N,\varepsilon}(\alpha)$ denote the least peak number of persistent labels achieving total-variation error at most ε on every seed, word and query; Definition 2.1 is precise. Set

$$b_\alpha = \inf_{l \geq 1} l \|l\alpha\|_{\mathbb{R}/\mathbb{Z}}.$$

We call α *badly approximable* when $b_\alpha > 0$; equivalently its continued-fraction partial quotients are bounded.

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Theorem 1.1 (Sharp hidden width). *Let α be badly approximable and $0 \leq \varepsilon < 1/20$ be fixed. In the clocked atomic stochastic-row model,*

$$(2) \quad W_{N,\varepsilon}(\alpha) = \Theta_{\alpha,\varepsilon}(N^{1/3}).$$

More explicitly, put $\kappa = 1/10 - 2\varepsilon$. Every realization of peak at most K satisfies

$$(3) \quad N < (2K - 1) + \frac{(2K - 1)^3}{b_\alpha^2} \log \frac{1}{\kappa}.$$

The upper bound is exact, uses two command matrices independent of the epoch, and has a decoder depending on N . At the golden angle $\alpha = (\sqrt{5} - 1)/2$ and zero error,

$$(4) \quad \max\{3, (N/168)^{1/3}\} \leq W_{N,0}(\alpha), \quad W_{N,0}(\alpha) < 2 \max\{5, (40N)^{1/3}\}.$$

In fact $W_{N,0} \geq 3$ and $W_{N,0} > (N/168)^{1/3}$ separately.

None of the constants is optimized. The conclusion identifies the order of width, not the exact integer optimum or its leading constant. In persistent bits it gives $(\log_2 N)/3 + O_{\alpha,\varepsilon}(1)$. This is not a uniform space bound charging program description or exact real arithmetic.

The predecessor Fourier argument gives $\Omega(N^{1/5})$ for the same unrestricted hidden-state class [15]. Its conditional phase amplitude is the starting point here. The improvement does not come from changing a trigonometric-kernel constant. Instead we choose correlated commands *inside* fresh independent packets. A packet of $2k - 1$ commands produces $2k$ separated phase increments. Whatever hidden mechanism is used inside it, a k -label endpoint loses a proportion of amplitude of order k^{-2} . Since a wordwise simulator must pass this input distribution, multiplication over disjoint packets gives the cube-root converse. This also controls the number of low-width cuts without bounding the remaining widths.

The underlying model is a nonuniform probabilistic ordered read-once branching program, with two terminal affine readouts. Section 7 compares the result with the Fourier-growth theorems of Reingold–Steinke–Vadhan, Steinke–Vadhan–Wan and Lee–Pyne–Vadhan [1, 2, 3]. Our packet argument is a hidden-compression estimate, not a theorem about pseudorandom generators or Boolean Fourier growth. The circle-walk literature [5] describes the uncompressed walk; conditional averaging in a small register is an additional operation.

Notation	Meaning
$K_t, W_{N,\varepsilon}$	Available labels at cut t ; optimum peak.
q_t	Expected norm of a conditional past-orbit mean, used only in the proof.
$s_B(\alpha)$	$\min_{1 \leq l \leq B} \ l\alpha\ _{\mathbb{R}/\mathbb{Z}}$.
$\Gamma_k(\mu)$	Worst starting-direction k -center orbit distortion.
$\mathcal{E}_N(\mathbf{K})$	Minimum actual terminal row error for a profile.
$\mathcal{D}(E)$	Accumulated local deficiency certificate of a diagram.

Contribution and provenance. The packet contraction, sharp unrestricted order, nonuniform packet budget and compact-group applications are the new results. The conditional-amplitude variable is inherited from [15]; the resonant-polygon upper construction is inherited from [14] and is proved again here. Jensen’s inequality, sphere quantization, continued fractions and finite Blackwell garbling are classical tools. The earlier Fourier theorem is preserved, with its angular estimate sharpened in Appendix A. No accumulated archive is needed to verify the new proofs in this article.

2. THE EXPERIMENT AND ITS REALIZING MACHINES

State distributions are row vectors and $\Delta_k = \{e \in \mathbb{R}^k : e \geq 0, e\mathbf{1} = 1\}$.

Definition 2.1 (Clocked stochastic realization). For a specified N , a realizing machine has finite state sets $[K_t]$ for $0 \leq t \leq N$, initialization rows $E_x \in \Delta_{K_0}$, row-stochastic matrices $T_{t,c} \in \mathbb{R}_+^{K_t \times K_{t+1}}$ for $t < N$, $c \in \{0, 1\}$, and decoder means $d_j \in [-1, 1]^{K_N}$. Its conditional answer mean is

$$E_x T_{0,c_1} \cdots T_{N-1,c_N} d_j.$$

Its error is one half the maximum absolute difference from (1), over all seeds, command words and queries. The profile counts *available* labels, including padding. Its peak is $\max_{0 \leq t \leq N} K_t$, or the maximum of this number and two if a binary answer is retained at an additional output cut. $W_{N,\varepsilon}(\alpha)$ is the least such peak at error at most ε .

Only the updated register persists. An earlier random seed, a transcript, an input-dependent clock or any retained protocol variable must be in that register. The current input is consumed in its update; seed and query are supplied just once. The epoch is external and the complete row table may depend on N . All rows, including unused rows, are normalized. Exact atomic stochastic rows and read-only tables are allowed; row construction, arithmetic and table length are not part of this width. A self-paced finite fair-bit compiler for rational rows has additional workspace and is a different resource statement [12].

The commands are externally specified inputs. No independence or fairness is part of the simulation task. A lower bound may impose any distribution on command words, including one chosen after the available profile is known. That distribution is used in the proof and is not a persistent seed supplied to the machine. When counting states under a chosen test law, zero-probability labels may be removed; the conclusions in terms of available labels remain valid.

Proposition 2.2 (Separate positive minimum). *At every cut after initialization the exact normalized residual rank and the separate positive minimum of (1) are three. The latter permits arbitrary finite registers at the other cuts.*

Proof. The predictive vectors are $R_\alpha^a x/10$. Idle prefixes from the four seeds span a plane, and two coordinate queries after an idle suffix separate it. Normalization adds the constant row, giving rank three. The triangle with vertices $(1/2, 0)$ and $(-1/2, \pm 1/2)$ contains the seed disk of radius $\sqrt{2}/10$, since its side distances from the origin are $1/2$ and $1/(2\sqrt{5})$. Its vertices have norm at most $1/\sqrt{2}$, so all rotated continuations give valid binary laws. At the chosen cut decompose the actual predictive vector in this triangle. Before it retain the finite raw history; afterwards retain the triangle label and the remaining active-command count. This is a complete finite machine attaining three at that cut. Rank gives the lower bound. This separate-cut construction is inherited from [13, 14]. $\square$

The statement does not require a hidden basis state to possess a predictive vector. Indeed the converse below never projects a hidden simplex, or bounds the number of vertices in such a projection.

3. ORBIT QUANTIZATION UNDER STOCHASTIC COMPRESSION

We first state the probabilistic argument in a real finite-dimensional Hilbert space H . A complex unitary representation is covered by taking its real inner product. Let $\mathbb{S}(H)$ be the unit sphere and let μ be a probability measure on a compact group acting orthogonally on H , with action U_g .

Definition 3.1 (Orbit distortion). For $k \geq 1$ set

$$(5) \quad \Gamma_k(\mu) = \inf_{v, u_1, \dots, u_k \in \mathbb{S}(H)} \int \min_{i \leq k} (1 - \langle U_g v, u_i \rangle) d\mu(g).$$

It depends only on the representation and the law of a command packet, not on a proposed stochastic realization. It lies in $[0, 1]$.

The upper bound one follows by using the direction of $\int U_g v d\mu(g)$ as one of the centers, with any center if that integral vanishes. The infimum is attained by compactness and continuity. An invariant direction may make the coefficient zero; positivity is a geometric property to be established, not an automatic conclusion for all representations.

Lemma 3.2 (One-packet contraction). *Let $Y \in H$ be integrable, let S be a finite register, and let G be independent of (Y, S) with law μ . Suppose the new register S' has at most k positive-probability values and its conditional law, given (Y, S, G) , depends only on (S, G) . Then*

$$(6) \quad \mathbb{E} \|\mathbb{E}[U_G Y \mid S']\| \leq (1 - \Gamma_k(\mu)) \mathbb{E} \|\mathbb{E}[Y \mid S]\|.$$

Proof. Write $z_s = \mathbb{E}[Y \mid S = s]$, $p_s = \mathbb{P}(S = s)$ and $T_g(s, r) = \mathbb{P}(S' = r \mid S = s, G = g)$. Conditional independence gives

$$p'_r z'_r = \sum_s p_s \int T_g(s, r) U_g z_s d\mu(g), \quad z'_r = \mathbb{E}[U_G Y \mid S' = r].$$

Choose $u_r = z'_r / \|z'_r\|$ when the latter is nonzero and any unit vector otherwise. Thus

$$\begin{aligned} \sum_r p'_r \|z'_r\| &= \sum_s p_s \int \sum_r T_g(s, r) \langle U_g z_s, u_r \rangle d\mu(g) \\ &\leq \sum_{s: z_s \neq 0} p_s \|z_s\| \int \max_r \langle U_g(z_s / \|z_s\|), u_r \rangle d\mu(g) \\ &\leq (1 - \Gamma_k(\mu)) \sum_s p_s \|z_s\|. \end{aligned}$$

Zero centroids cause no loss of an identity: their inner product with any chosen direction is zero. The maximization only relaxes the actual possibly randomized assignment. In particular, neither the old register nor the computation inside the packet is required to have k states. $\square$

Theorem 3.3 (Representation-valued packet budget). *Let Y_0 be a fixed unit vector. Each command packet is sampled independently of the earlier process, with its full word law fixed in advance. Write G_i for its product and μ_i for the induced law. Between packets one may insert fixed commands. A finite stochastic register processes the commands in their prescribed order, with arbitrary cut-dependent rows. Let k_i bound its positive-probability states at the end of packet i . If a terminal function $d(S_N)$ satisfies $\|d\| \leq B$ and*

$$|\mathbb{E} \langle Y_N, d(S_N) \rangle| \geq B\kappa > 0,$$

then

$$(7) \quad \kappa \leq \prod_i (1 - \Gamma_{k_i}(\mu_i)), \quad \sum_i \Gamma_{k_i}(\mu_i) \leq \log(1/\kappa).$$

Here Y evolves by the same ordered representation products as the commands. The statement also applies to a complex scalar correlation with absolute value, using $q_N \geq \kappa$ as its terminal hypothesis.

Proof. At each boundary put $q = \mathbb{E}\|\mathbb{E}[Y \mid S]\|$. A whole packet has one stochastic input/output kernel conditional on its starting state and its command word. Its internal rows and all internal hidden states are integrated into that kernel. The fresh packet is independent of the past, so Lemma 3.2 applies. A fixed command can only decrease q , by conditional Jensen and preservation of norm by its action. Initially $q = 1$. At the terminal cut, $|\mathbb{E}\langle Y_N, d(S_N) \rangle| \leq Bq_N$. Multiplication proves the first assertion; $1 - u \leq e^{-u}$ proves the second. If a coefficient is one, positive terminal correlation is impossible, consistent with the first assertion. $\square$

The products G_i need not commute. The packet endpoints need not be consecutive cuts. Most importantly, the commands *inside* a packet need not be independent. Only its independence from the earlier process is used in the conditional-mean identity. This is what permits a stronger converse than a fair one-command test.

Proposition 3.4 (A small-ball criterion). *For $r > 0$ define*

$$A_\mu(r) = \sup_{v, u \in \mathbb{S}(H)} \mu\{g : \|U_g v - u\| < r\}.$$

Then

$$(8) \quad \Gamma_k(\mu) \geq \frac{r^2}{2} (1 - kA_\mu(r))_+.$$

If for every v the packet is uniform on m orbit points separated by Euclidean distance at least σ , then $\Gamma_k(\mu) \geq (1 - k/m)_+ \sigma^2/8$.

Proof. Outside the union of k radius- r balls the squared distance to each center is at least r^2 . Use $1 - \langle v, u \rangle = \|v - u\|^2/2$ and the union bound. For separated points, each open ball of radius $\sigma/2$ contains at most one, so $A_\mu(\sigma/2) \leq 1/m$. $\square$

4. THE SHARP BOUND FOR BINARY ROTATION COMMANDS

Put $\zeta = e^{2\pi i \alpha}$ and

$$(9) \quad s_B(\alpha) = \min_{1 \leq l \leq B} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}}.$$

For irrational α this number is positive at every finite B . Given an integer $k \geq 1$, set $B = 2k - 1$ and choose J uniformly from $\{0, \dots, B\}$. The associated packet is the length- B command word

$$(10) \quad 1^J 0^{B-J}.$$

The command count is uniform on $2k$ possibilities; the physical packet still consists of B ordinary binary commands. Its random choice is not revealed as an additional input.

Lemma 4.1 (Separated packets). *The circle action of the packet (10) has*

$$(11) \quad \Gamma_k(\mu_B) \geq s_B(\alpha)^2.$$

Consequently it multiplies the conditional phase amplitude by at most $1 - s_B(\alpha)^2$ if its endpoint register has at most k states.

Proof. Write $s = s_B(\alpha)$. For any starting direction the $2k$ phase points differ pairwise by at least s in normalized circle distance. Each open arc of radius $s/2$ contains at most one of them. For any k center directions, at least k packet points are therefore at circle distance at least $s/2$ from every center. Their loss $1 - \cos(2\pi\theta)$ is at least $1 - \cos(\pi s)$. Since $0 \leq s \leq 1/2$ and $\sin(\pi s/2) \geq s$,

$$\frac{1}{2}(1 - \cos(\pi s)) = \sin^2(\pi s/2) \geq s^2.$$

Average over the $2k$ points and minimize over centers and initial phase. Lemma 3.2 proves the last assertion. This argument concerns actual hidden-state centroids, not their observable lift vertices. $\square$

Lemma 4.2 (Terminal calibration for every test law). *Fix seed $x = (1, 1)$. Under any command-word distribution independent of the machine's private randomness, a realization with error $\varepsilon < 1/20$ obeys*

$$(12) \quad q_N := \mathbb{E}|\mathbb{E}[\zeta^{L_N} \mid S_N]| \geq \kappa := 1/10 - 2\varepsilon.$$

Proof. Set $d = d_1 + id_2$, so $|d| \leq \sqrt{2}$. For each word the target conditional mean of $d(S_N)$ is $(1+i)\zeta^{L_N}/10$. A binary row-TV error ε changes each coordinate mean by at most 2ε , hence the complex mean by at most $2\sqrt{2}\varepsilon$. Multiply by ζ^{-L_N} and average to get

$$|\mathbb{E}[\zeta^{-L_N} d(S_N)]| \geq \sqrt{2}(1/10 - 2\varepsilon).$$

Conditioning on S_N bounds the left side by $\sqrt{2}q_N$. Only correctness at the final prescribed length was used. $\square$

Theorem 4.3 (Finite packet and occupation bounds). *For an arbitrary irrational α , a machine of error $\varepsilon < 1/20$ and available-label profile $(K_t)_{t=0}^N$ satisfies, for every $k \geq 1$ and $B = 2k - 1$,*

$$(13) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B}{s_B(\alpha)^2} \log \frac{1}{\kappa}.$$

In particular, a peak- K machine has

$$(14) \quad \left\lfloor \frac{N}{2K-1} \right\rfloor s_{2K-1}(\alpha)^2 \leq \log \frac{1}{\kappa}.$$

More generally, for pairwise disjoint packets ending at cuts t_i , of lengths $B_i = 2k_i - 1$, with $K_{t_i} \leq k_i$, one has

$$(15) \quad \prod_i (1 - s_{B_i}(\alpha)^2) \geq \kappa, \quad \sum_i s_{B_i}(\alpha)^2 \leq \log \frac{1}{\kappa}.$$

Proof. On the chosen disjoint intervals use independent packets (10); set all other commands to zero. Fresh packet independence, Lemma 4.1 and Theorem 3.3 apply. Lemma 4.2 supplies the final correlation for this particular test law. This proves (15), and consecutive length- $2K - 1$ packets prove (14).

For (13), disregard low-width endpoints before B , of which there are at most $B - 1$. Select the first remaining such endpoint t , take the interval $(t - B, t]$, and repeat at the first low endpoint at least B epochs later. The selected intervals are disjoint. Each selected endpoint removes at most B low endpoints, since there are only B integer epochs in $[t, t + B - 1]$. If m intervals are selected, the total number of low endpoints is at most $B - 1 + Bm$. The second inequality in (15) gives $ms_B(\alpha)^2 \leq \log(1/\kappa)$.

The choice of intervals uses only the available profile, not the realized state or past command. It defines a legitimate external test distribution for this machine. No width bound is used inside a selected packet or at any unselected cut. $\square$

Proof of the lower bound in Theorem 1.1. For badly approximable α , $s_B(\alpha) \geq b_\alpha/B$. Equation (14) and $N < B(\lfloor N/B \rfloor + 1)$ prove (3). Since $2K - 1 \leq 2K$ and $K \leq K^3$,

$$(16) \quad W_{N,\varepsilon}(\alpha) > \left(\frac{N}{2 + 8b_\alpha^{-2} \log(1/\kappa)} \right)^{1/3}.$$

The corresponding occupation bound is $\#\{1 \leq t \leq N : K_t \leq k\} \leq 2k - 2 + 8b_\alpha^{-2} k^3 \log(1/\kappa)$.

For the golden angle, let $\alpha' = -(1 + \sqrt{5})/2$ and let p be nearest to $l\alpha$. Then

$$(l\alpha - p)(l\alpha' - p) = p^2 + lp - l^2 \in \mathbb{Z} \setminus \{0\}, \quad |l\alpha' - p| \leq \sqrt{5}l + 1/2 < 3l.$$

Thus $b_\alpha \geq 1/3$. At zero error, $2 + 72 \log 10 < 168$, proving the stated bound. The elementary inequality $\log 10 < 83/36$ can, for example, be checked by the positive exponential series. The rank-three bound follows from Proposition 2.2. $\square$

4.1. Matching realization. We reproduce the resonant upper construction from [14, 15] to make the match self-contained. Let p/q be a convergent of α with $q \geq \max\{5, (40N)^{1/3}\}$. Put

$$a = \pi/q, \quad \delta = 2\pi(\alpha - p/q), \quad \lambda = \frac{\cos(a - |\delta|)}{\cos a}, \quad \omega = \frac{\sin |\delta|}{\lambda \sin(2a)}, \quad u_s = (\cos(2\pi s/q), \sin(2\pi s/q)).$$

Subscripts on u_s are modulo q . Convergence gives $|\delta| < a$, so $\lambda \geq 1$ and $0 \leq \omega \leq 1$.

Proposition 4.4 (Exact resonant upper bound). *The experiment has an exact q -state realization. Its two command matrices are independent of the epoch, although its decoder depends on N . For badly approximable α , one may take $q = O_\alpha(N^{1/3})$.*

Proof. Associate to label s at time t the vector $r_t u_s$, $r_t = \lambda^t/4$. Define

$$T_0(s, \cdot) = \lambda^{-1} \delta_s + (1 - \lambda^{-1}) \text{Unif}[q], \quad T_1(s, \cdot) = (1 - \omega) \delta_{s+p} + \omega \delta_{s+p+\text{sgn } \delta}.$$

When $\delta = 0$ take $\omega = 0$. The chord between the two indicated active vertices meets the ray at angle $2\pi(s/q + \alpha)$ at radius $1/\lambda$. Its second coordinate gives exactly the formula for ω . Thus the expected next associated vector is $r_t u_s$ for an idle command and $R_\alpha r_t u_s$ for an active command. Furthermore

$$\log \lambda = \int_{a-|\delta|}^a \tan v \, dv \leq |\delta| \tan a < 4\pi^2/q^3 < 40/q^3,$$

since $q \geq 5$. Hence $r_N \leq e/4 < 1$. The initial polygon has inradius $\cos(\pi/q)/4 > \sqrt{2}/10$; initialize by any convex decomposition of $x/10$. The final decoder has means $r_N e_j^T u_s \in [-1, 1]$. Induction proves exactness for all words. Bounded partial quotients bound consecutive convergent-denominator ratios. For the golden angle the first denominator above a threshold $L \geq 5$ is less than $2L$, proving (4). $\square$

Corollary 4.5 (Hidden, vector and fixed-length common-row widths). *For a badly approximable angle, exact clocked hidden width, exact clocked vector-state width, and exact fixed-length width with common command matrices are all $\Theta_\alpha(N^{1/3})$.*

Proof. The first class contains the other two, while Proposition 4.4 belongs to their intersection. Apply the unrestricted lower bound. $\square$

An unchanged decoder correct after every earlier stopping length is a stronger, anytime specification; no such assertion is used here.

4.2. Other arithmetic. If $\|l\alpha\| \geq bl^{-\tau}$ for all $l \geq 1$, the same proof gives $N < B + b^{-2}B^{2\tau+1} \log(1/\kappa)$ for $B = 2K - 1$. No matching claim is made under this weaker arithmetic hypothesis. For the irrational angle whose rotation has rational entries $\zeta = (3 + 4i)/5$, the nonzero Gaussian integer $(3 + 4i)^l - 5^l$ gives $s_B(\alpha) \geq 5^{-B}/(2\pi)$. Consequently

$$N < B + 4\pi^2 B 25^B \log(1/\kappa), \quad B = 2K - 1.$$

The Gaussian integer is nonzero because otherwise $6/5 = \zeta + \zeta^{-1}$ would be an algebraic integer. This is an explicit logarithmic converse, not a proof that this angle is badly approximable. The earlier rational counter realization and its finite-coin accounting remain unchanged.

5. COMPACT COMMAND GROUPS

Theorem 3.3 applies to any orthogonal or unitary representation, including noncommuting commands. The following two consequences distinguish a finite command alphabet from a genuinely compact command alphabet.

Corollary 5.1 (Characters of compact groups). *Let a finite command alphabet in a compact group contain the identity and g , and let χ be a continuous one-dimensional unitary character with $\chi(g) = e^{2\pi i\alpha}$, α irrational. Any wordwise finite-register experiment whose terminal specification forces a conditional χ -amplitude at least $\kappa > 0$ satisfies (13) and (15) on these commands. In particular, a badly approximable character angle forces $\Omega(N^{1/3})$ hidden width.*

Proof. Use packets $g^J e^{B-J}$. Their character images are exactly those in Lemma 4.1. The other commands may be assigned probability zero by the test law. Apply Theorem 3.3 on $\mathbb{C}$. $\square$

This includes higher-dimensional tori through their characters. It is a lower bound for a selected observed character, not a multiplicative claim about simultaneous independent outputs of several characters. The next result uses a different orbit geometry rather than a character.

Fix an integer $d \geq 2$ and a unit vector $v \in \mathbb{R}^d$. Consider the separate experiment with commands $g_t \in \text{SO}(d)$, supplied as exact atomic matrices, and final coordinate query $j \in \{1, \dots, d\}$. Its specified mean is

$$(17) \quad \mathbb{E}[B \mid g_1, \dots, g_N, j] = \frac{1}{10} e_j^T g_N \cdots g_1 v.$$

A realizing finite-state machine has Borel stochastic rows $T_{t,g}$. Its width still counts only labels; access to this compact input and its row function is part of this model, not an implementation by finite binary input symbols. All terminal probabilities lie in $[9/20, 11/20]$.

Lemma 5.2 (Sphere quantization). *For normalized surface measure on $\mathbb{S}^{d-1}$ there are constants $c_d, C_d > 0$ such that the orbit distortion of Haar measure on $\text{SO}(d)$ satisfies*

$$\Gamma_k(\text{Haar}) \geq c_d k^{-2/(d-1)}.$$

For every $0 < \rho \leq 1/2$ there is a finite set $V \subset \mathbb{S}^{d-1}$ of at most $C_d \rho^{-(d-1)}$ points whose convex hull contains the Euclidean ball of radius $\cos \rho$.

Proof. Haar rotation sends every unit vector to normalized surface measure. A Euclidean radius- r cap, $0 < r \leq 1$, has angular radius $2 \arcsin(r/2) \leq \pi r/2$. Polar integration bounds its probability by $A_d r^{d-1}$ for some $A_d \geq 1$: its integral is proportional to $\int_0^{2 \arcsin(r/2)} \sin^{d-2} u \, du$. Take $r = (2A_d k)^{-1/(d-1)}$ in Proposition 3.4. The bound follows with $c_d = (2A_d)^{-2/(d-1)}/4$.

For the second assertion choose a maximal set with pairwise angular distance greater than ρ . The angular radius- $\rho/2$ caps are disjoint and each has surface mass at least $a_d \rho^{d-1}$, by the same polar integral and $\sin u \geq 2u/\pi$ on $[0, \pi/2]$. Thus the set has at most $a_d^{-1} \rho^{-(d-1)}$ points. Maximality makes it a ρ -net. Its support function in every unit direction is at least $\cos \rho$, proving the ball containment. These elementary estimates are the usual spherical quantization and covering mechanisms; their role here is sequential compression. $\square$

Theorem 5.3 (Sharp width for all rotation commands). *For fixed $d \geq 2$ and $0 \leq \varepsilon < 1/(20\sqrt{d})$, the least hidden-state width for (17), in its compact-command model, satisfies*

$$(18) \quad W_{N,\varepsilon}^{\text{SO}(d)} = \Theta_{d,\varepsilon}(N^{(d-1)/2}).$$

Writing $\kappa_d = 1/(10\sqrt{d}) - 2\varepsilon$, every profile obeys

$$(19) \quad c_d \sum_{t=1}^N K_t^{-2/(d-1)} \leq \log(1/\kappa_d).$$

The exact upper construction has common command rows and an N -dependent terminal decoder.

Proof. Use independent Haar commands as the test law and set $Y_t = g_t \cdots g_1 v$. Combine the coordinate decoder means into a vector $d(S_N)$ of norm at most $\sqrt{d}$. Wordwise approximation implies

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq 1/10 - 2\sqrt{d}\varepsilon, \quad q_N \geq \kappa_d.$$

Theorem 3.3 and Lemma 5.2, one command per packet, give (19) and the lower bound.

For the upper bound take $\rho = 1/(2\sqrt{N})$ and a net V as in Lemma 5.2. Put $P = \text{conv } V$, $\lambda = (\cos \rho)^{-1}$, and $r_t = \lambda^t/4$. A label $u \in V$ has associated vector $r_t u$. For any command g , the vector gu/λ lies in $(\cos \rho)\mathbb{B}^d \subset P$. Fix a Borel convex decomposition of every point of P into V , and use the coefficients of gu/λ as the command row from label u . Such decompositions may be chosen by a finite triangulation of P using its vertices, with ties between simplices resolved in a fixed order. The resulting rows are Borel, normalized, and independent of t .

The expected next associated vector is exactly $g(r_t u)$. Since $-\log \cos \rho \leq \rho^2$ for $\rho \leq 1/2$, $r_N \leq e^{1/4}/4 < 1$. Also $v/10 \in r_0 P$, since $\cos(1/2)/4 > 1/10$. Initialize by its chosen convex decomposition and decode with means $r_N e_j^T u$. This realizes (17) exactly and uses at most $C_d(2\sqrt{N})^{d-1}$ labels (and two output labels, absorbed by the bound). No vectors are retained at runtime. $\square$

For $d \geq 3$ the commands here are noncommuting. For $d = 2$ the order $N^{1/2}$ concerns all circle rotations as inputs, while Theorem 1.1 has the smaller fixed alphabet $\{I, R_\alpha\}$ and order $N^{1/3}$. These statements concern different command sets. The compact theorem is not a claim of that exponent for an arbitrary finite generating set, or an uncharged approximation of a real command by a finite word.

6. ACTUAL TERMINAL ERROR AND LOCAL DIAGRAM CERTIFICATES

For a prescribed profile $\mathbf{K}$, define $\mathcal{E}_N(\mathbf{K})$ as the minimum uniform terminal row-TV error in Definition 2.1. The minimum exists: the row simplices, initialization rows and decoder intervals form a compact set, and the maximum of finitely many wordwise errors is continuous. Thus this is an intrinsic final simulation objective, not a sum of intermediate defects.

For $t \geq 1$ put $B_t = 2K_t - 1$. Call $(t - B_t, t]$ admissible when $B_t \leq t$. Let

$$(20) \quad \mathfrak{B}_\alpha(\mathbf{K}) = \max_{\substack{\text{pairwise disjoint} \\ \text{admissible intervals}}} \sum_i s_{B_{t_i}}(\alpha)^2.$$

This finite weighted-interval problem depends only on the profile and angle. Empty selections give zero. If $F_0 = 0$, its exact recurrence is

$$F_t = \begin{cases} \max\{F_{t-1}, F_{t-B_t} + s_{B_t}(\alpha)^2\}, & B_t \leq t, \\ F_{t-1}, & B_t > t. \end{cases} \quad \mathfrak{B}_\alpha(\mathbf{K}) = F_N.$$

The last chosen interval proves the recurrence. Computing the weights requires the stated angle representation; no bit-complexity claim for arbitrary exact real inputs is made.

Theorem 6.1 (Final-error obstruction). *For every irrational α and finite profile,*

$$(21) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{1}{10} - \exp\{-\mathfrak{B}_\alpha(\mathbf{K})\} \right]_+.$$

In particular, for peak at most K and $B = 2K - 1$,

$$(22) \quad \mathcal{E}_N(K) \geq \frac{1}{2} \left[\frac{1}{10} - \exp\{-\lfloor N/B \rfloor s_B(\alpha)^2\} \right]_+.$$

For a badly approximable angle, every sequence $K = o(N^{1/3})$ has $\liminf \mathcal{E}_N(K) \geq 1/20$.

Proof. Choose the disjoint intervals maximizing (20) and put independent separated packets on them, idle commands elsewhere. Every machine of that profile has terminal amplitude at most $\exp\{-\mathfrak{B}_\alpha(\mathbf{K})\}$, by Lemma 3.2 and Lemma 4.1. Terminal calibration gives $1/10 - 2\varepsilon \leq q_N$ whenever its left side is positive. Rearrange and then minimize over machines. Consecutive equal-length packets prove (22); the estimate $s_B \geq b_\alpha/B$ proves the asymptotic assertion. $\square$

This conclusion is about the actual terminal error, so possible cancellation of intermediate approximation errors cannot evade it. It does not give the exact value of $\mathcal{E}_N$ at every finite profile. In particular the limiting lower constant $1/20$ is not claimed optimal.

6.1. Transfer to executable diagrams. For finite row-stochastic experiments E, F indexed by the same finite set of prefixes, use the classical one-sided deficiency

$$(23) \quad \delta(E, F) = \min_{T \text{ stochastic}} \max_h \text{TV}((ET)_h, F_h).$$

The common row T cannot depend on h . The zero-defect case is Blackwell garbling [10, 11]; composition contracts total variation.

Let E_t specify a proposed state distribution for each actual input prefix at cut t . For every next command, choose a common update approximating the corresponding child rows, and likewise choose each terminal query decoder. Let ϵ_t be the worst row error of that update over prefixes and commands, and ϵ_{out} the worst terminal decoder error. If the initial rows are exact, put

$$\mathcal{D}(E) = \sum_{t < N} \epsilon_t + \epsilon_{\text{out}}.$$

Minimizing each one-step error gives (23) at that step. This is an *accumulated local deficiency certificate*, not a formula for the optimal final error.

Corollary 6.2 (Diagram obstruction). *Every such diagram of profile $\mathbf{K}$ satisfies the right side of (21) as a lower bound on $\mathcal{D}(E)$.*

Proof. Execute the selected common kernels. At each fixed prefix, induction and TV contraction bound the discrepancy from E_t by the sum of preceding ϵ_s . Append the common terminal decoder to obtain actual row error at most $\mathcal{D}(E)$. If this sum exceeds one, the trivial error bound one suffices. Thus $\mathcal{D}(E) \geq \mathcal{E}_N(\mathbf{K})$; apply Theorem 6.1. $\square$

The certificate can overestimate the achieved final error, since local errors may cancel. The corollary does not deny cancellation, minimize $\mathcal{D}$ over all diagrams, or assert equality of the two objectives. The new lower bound already applies before any diagram is introduced.

7. COMPARISON OF MODELS AND CONCLUSIONS

7.1. Probabilistic read-once branching programs. After fixing the seed and padding layers to a common width, the product in Definition 2.1 is a stochastic ordered read-once branching program: layer t reads command c_t once and applies T_{t-1, c_t} . This is the randomized-transition extension of the standard nonuniform model in [1, Section 1.1]; see also [4]. Width is the maximum layer size, whereas total size counts all layers. Our theorem is a width theorem, not a succinct uniform program-size or bit-operation theorem.

For a chosen query j , appending an accepting Bernoulli row with probability $(1 + d_j(s))/2$ turns the real readout into ordinary randomized acceptance with no larger width than $\max(K, 2)$. Combining the two means into $d_1 + id_2$ does not enlarge the retained alphabet. This combination is an identity between mean functions, not a requirement to sample both query answers jointly. Thus real readouts and the word “transducer” do not by themselves separate our problem from branching programs.

For a fixed initial row p and terminal column d , put $F(c) = pT_{0,c_1} \cdots T_{N-1,c_N}d$. Its Boolean-input Fourier coefficient at $A \subseteq [N]$ is exactly

$$(24) \quad \widehat{F}(A) = p \prod_{t=1}^N \frac{T_{t-1,0} + (-1)^{\mathbf{1}_{\{t \in A\}}} T_{t-1,1}}{2} d.$$

This follows by multiplying independent sums over input bits in their original order. It is a matrix-product Fourier coefficient, distinct from a harmonic of the conditional phase measure used in [15]. The present proof instead uses the nonlinear quantity $\mathbb{E}|\mathbb{E}[\zeta^{L_t} | S_t]|$ at packet boundaries.

Reingold–Steinke–Vadhan [1, Theorem 1.4] bound Boolean level- l Fourier mass by $(2w^2)^l$ for regular width- w programs. Steinke–Vadhan–Wan [2] establish width-three Fourier-growth bounds and use them for pseudorandomness. Lee–Pyne–Vadhan [3, Definitions 3–5 and Theorem 6] refine the regular-program bounds, including unbounded width and restrictions on accepting states; their Section 1.2.3 discusses generalized group products. These results concern Fourier growth or fooling classes of computations. They do not state the terminal-correlation packet inequality (6) or its nonuniform occupation consequence.

Regularity means preservation of a uniform layer distribution under a fair next input. It is not implied by row stochasticity: the matrix all of whose rows put mass one on a single label is legal here and destroys uniformity. Consequently a regular/permutation theorem cannot be invoked without its hypothesis. Conversely, some bounds for general deterministic programs extend to stochastic programs by resolving all private rows into a convex combination of deterministic tables and using convexity. We therefore make no claim that private randomness alone prevents use of the earlier literature. The extra statement proved here is the multiplicative loss forced by a separated command packet, for arbitrary starting conditional centroids and arbitrary endpoint assignments. It gives the matching growing-width law without a regularity assumption and without an estimate of the Fourier mass in (24).

7.2. Circle walks and the choice of test law. For independent Bernoulli commands of bias p , the uncompressed phase has harmonic multiplier

$$a_l = 1 - p + pe^{2\pi i l \alpha}, \quad |a_l|^2 = 1 - 4p(1-p)\sin^2(\pi l \alpha).$$

These are classical circle-walk quantities. Berkes–Borda [5] relate convergence and limit theorems for circle walks to rational approximation, including the transition along convergent denominators. That analysis concerns the uncompressed walk, not the information retained by an additional hidden register.

The earlier Fourier budget [15] combined this damping, common conditional-amplitude loss, and a positive finite-atom Fourier witness. It gave a fifth-root unrestricted converse. Our test law is different: its bits are correlated inside a packet so that the packet count is uniform, and different packets are independent. We are not asserting a cube-root occupation theorem for the fair-command law itself. Wordwise correctness allows the new law; Lemma 3.2 measures its compression directly. The packet costs $O(k)$ commands and loses $\Omega(k^{-2})$ amplitude, explaining the cube-root order. The angular averaging estimate remains valid with its sharp universal coefficient, proved in Appendix A; improving that coefficient’s quadratic order is not needed for the new result.

7.3. Positive realization and experiment comparison. Invariant convex sets, positive stochastic realizations, and irrational rotation obstructions for one repeated positive matrix belong to classical realization theory [6, 7, 8, 9]. The present lower bound permits unrelated matrices at every layer and a new machine for each horizon; it is not a peripheral-eigenvalue argument for a stationary realization. The upper construction is inherited from [14, 15] and has

been proved again in Proposition 4.4. What is new in the main theorem is its match against all hidden states, including under fixed row error.

Finite Blackwell garbling and one-sided deficiency are classical [10, 11]. Theorem 6.1 bounds the actual terminal optimization. Corollary 6.2 then bounds an accumulated local certificate; it does not identify that certificate with the optimized terminal distance. Orbit quantization, sphere-cap bounds and polytope nets in Section 5 are classical geometric mechanisms. Their representation-valued sequential loss is stated and proved explicitly in Theorem 3.3.

7.4. Conclusions and precise boundaries. The hidden and vector-state rotation widths have the same cube-root order for each badly approximable angle and fixed nonzero signal at sufficiently small fixed error. The finite constants are proof constants, not optimized thresholds or exact integer minima. The packet budget also controls selected low-width endpoints and actual terminal error. For compact rotation commands, a distinct matched sphere-dimension law follows from the same contraction theorem.

No sharp order is asserted for every irrational angle, for every finite noncommuting command set, or for a regime in which signal and permitted error vary with the horizon. Optimizing informative action choice is not the external-command problem studied here. Atomic computable-real rows are not finite fair-bit algorithms. No exact minimum dynamic Blackwell distance or independent analytic foundation for an unrelated model is inferred from these results. The original manuscript and its historical arguments remain separately available, with their hypotheses unchanged.

APPENDIX A. THE SHARP HOMOGENEOUS ANGULAR COEFFICIENT

The earlier Fourier proof remains useful when the external test law is required to have independent single commands. Its elementary angular estimate admits the following sharpened statement. This also resolves the coefficient issue in the preceding report without changing any previous published source.

Lemma A.1 (Homogeneous angular averaging). *Let $f_l(re^{i\theta}) = re^{il\theta}$ for $r > 0$ and $f_l(0) = 0$. For every integrable complex Z and integer $l \geq 1$,*

$$(25) \quad |\mathbb{E}f_l(Z) - f_l(\mathbb{E}Z)| \leq (l^2 - 1)(\mathbb{E}|Z| - |\mathbb{E}Z|).$$

For $l \geq 2$ the coefficient $l^2 - 1$ cannot be decreased uniformly over all such mixtures. At $l = 1$ both sides vanish.

Proof. The case $l = 1$ is linear. For $l \geq 2$, if $\mathbb{E}Z = 0$, the left side is at most $\mathbb{E}|Z|$ and $l^2 - 1 \geq 1$. Otherwise rotate the variable so $\mathbb{E}Z = \rho > 0$, and write $Z = re^{i\theta}$. For $u = e^{i\theta}$,

$$u^l - 1 - l(u - 1) = (u - 1) \sum_{j=0}^{l-1} (u^j - 1).$$

Since $|u^j - 1| \leq j|u - 1|$, the modulus of this remainder is at most $l(l - 1)(1 - \cos \theta)$. Retaining this coefficient gives

$$|u^l - \cos \theta - il \sin \theta| \leq \{l(l - 1) + (l - 1)\}(1 - \cos \theta) = (l^2 - 1)(1 - \cos \theta).$$

Multiply by r and average. The identities $\mathbb{E}(r \sin \theta) = 0$ and $\mathbb{E}(r \cos \theta) = \rho$ prove (25). Rotation covariance restores the general case. For sharpness take $Z = e^{i\theta}$ and $e^{-i\theta}$ with equal probabilities, and let $\theta \downarrow 0$. The ratio is

$$\frac{|\cos(l\theta) - \cos \theta|}{1 - \cos \theta} \rightarrow l^2 - 1.$$

□

Thus a uniformly linear-in- l replacement is impossible for unrestricted conditional mixtures. The improvement to cube-root width in this paper does not require such an estimate: it changes the test law to packets and measures the directly incurred geometric amplitude loss.

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